Re-analyzing the Effect of Right Heart Catheterization on ICU Mortality

A Doubly Robust Approach with E-value Sensitivity Analysis

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1. Research Question & Data

Causal Query: What is the effect of receiving right heart catheterization (RHC) within 24h on the 30-day mortality of adult ICU patients?

Study Variables:

- **Treatment (A):** Binary indicator for RHC within 24 hours.
- **Outcome (Y):** 30-day all-cause mortality.
- **Covariates (L):** 62 baseline clinical and demographic variables.

Data Source:

- SUPPORT observational study ($n = 5,735$).
- 2,184 treated patients (38% of sample).

2. Causal Model

Key Structural Assumptions:

- Treatment assignment is driven by acute physiological derangement (e.g., shock, low BP), not baseline demographics.
- **Primary Confounders:** Day 1 vitals, APACHE III score, and Do Not Resuscitate (DNR) status.

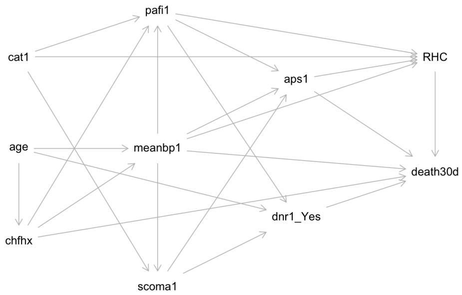

Figure: Proposed DAG identifying baseline variables confounding $A \to Y$

3. Identification Strategy

We identify the causal effect using the **backdoor adjustment criterion**.

Core Backdoor Adjustment Set (L):
Demographics + comorbidities + Day 1 physiology (DAG confounders).

Required Assumptions:

- **Consistency:** $Y = Y(A)$
- **Positivity:** $0 < P(A = 1 \mid L) < 1$
- **Ignorability:** $Y(a) \perp A \mid L$

Identification Formulas:
ATE:

$$E\{Y(1) - Y(0)\} = E_L\left[E(Y \mid A = 1, L) - E(Y \mid A = 0, L)\right]$$

ATT:

$$E\{Y(1) - Y(0) \mid A = 1\} = E_{L|A=1}\left[E(Y \mid A = 1, L) - E(Y \mid A = 0, L)\right]$$

Results: Average Treatment Effect on the Treated (ATT)

Estimator	Estimate	SE	95% CI Lower	95% CI Upper
Regression	0.060	0.013	0.036	0.088
IPW	0.047	0.014	0.019	0.074
PS Matching	0.041	0.017	0.006	0.074
Augmented IPW	0.049	0.014	0.022	0.075

Takeaway: *The unadjusted gap indicated a 7.5% mortality increase. After adjustment, estimates attenuate but remain consistently positive (∼4–6%).*

5. Diagnostics

Covariate Balance & Overlap:

- **Severe Imbalance:** Physiological indicators (APACHE, BP, PAO2) showed massive baseline imbalances (SMD > 0.4).
- **Demographics:** Age, race, and sex were relatively balanced at baseline (SMD < 0.1).
- **Conclusion:** Strong confounding by clinical indication, largely mitigated through our adjustment sets.

fig/balance-diagnostics-basic.png

6. Results & Interpretation

Main Finding: RHC is robustly associated with an increased 30-day mortality rate ($+4.9\%$ via Augmented IPW).

Subgroups (CATE): Harmful effects are amplified in severe cases:

- High-APACHE Score: $+10.5\%$
- Day-1 DNR Status: $+21.1\%$

Sensitivity Analysis (E-Values):

- Adjusted Risk Ratio $\approx 1.19 \implies$ **E-value** $\approx$ **1.67**.
- *Vulnerability:* The observed covariate `dnr1_Yes` has a Risk Ratio (~ 1.98) stronger than this threshold. Thus, the analysis is **not** robust to unmeasured confounding.

7. Broader Context

Methodological Alternatives:

- **Difference-in-Differences (DID):** Not applicable. The dataset lacks sufficient longitudinal pre- and post-treatment observations.
- **Instrumental Variables (IV):** Not applicable. No valid instrument exists, as RHC assignment is heavily determined by patient severity in the ICU.

Future Extensions:

- **Expert Causal Discovery:** Collaborate with cardiovascular surgeons to apply causal discovery algorithms, ensuring the DAG accurately reflects the complexities of acute ICU physiology.